

Weather Forecast Web Application

Explanation guide for ER Diagram & DFD (SYBSC-CA Mini Project — Ziya Shaikh)

1. DFD Level 0 — Context Diagram

What it shows: the whole system as a single process, exchanging data with two external entities — the **User** and the **Weather API**.

How to explain it out loud:

“This diagram shows my system's boundary. There are two external entities — User and Weather API. The User gives a city name to the system. The system sends a request to the Weather API to fetch that city's data. The API sends back a response, and the system then shows the result — weather info or an error — back to the User.”

Key point: at this level, the entire Flask application is treated as one black box — no internal detail yet.

2. DFD Level 1 — Detailed Diagram

What it shows: the black box from Level 0 broken into the four modules from the synopsis.

Process	What it does
1.0 City Search	Takes the city name from the user and validates it
2.0 Weather Fetch	Sends the validated city name to the Weather API and receives the response
3.0 Weather Display	If the city is valid, formats and shows the weather data to the user
4.0 Error Handling	If the city is invalid/misspelt, shows an error message to the user

How to explain it out loud:

“Now I break that black box into 4 processes, matching my 4 modules: City Search validates the city name the user enters. Weather Fetch takes that validated name and calls the Weather API, then receives the response. From there, the flow splits into two paths — if the city is valid, it goes to Weather Display, which formats and shows the result. If the city is invalid, it goes to Error Handling, which shows an error message instead.”

Key point: Process 2.0 has two outgoing paths — a success path to 3.0 and a failure path to 4.0. That split is exactly what your synopsis calls the Error Handling Module.

3. ER Diagram — Data Model

What it shows: a possible data model, in case city searches are stored.

Entity	Purpose
CITY	Stores city name and country
SEARCH	One record per search — links to a city, records search time
WEATHER_DATA	The successful result of a search — temperature, humidity, wind speed
ERROR_LOG	Record of a failed search (invalid city)

How to explain it out loud:

“This diagram shows the data model — if I wanted to store city searches, these would be the entities. CITY stores the city name and country. SEARCH is created every time a user searches a city — it holds a reference to that city and the search time. WEATHER_DATA is the result of a successful search — temperature, humidity, wind speed. ERROR_LOG stores a record when a search fails, like an invalid city name.”

Relationship to mention: “One city can be searched many times — that’s a one-to-many relationship between CITY and SEARCH. And each search results in either weather data or an error, not both at the same time.”

If asked whether this database is actually implemented in the project: be honest that the synopsis does not mention a database — this ER diagram is a conceptual data model based on the City Search and Error Handling modules. If no database was actually built, you can say: “This is a conceptual data model; persistence is a possible future scope.”